

If I can **successfully ping the website's IP address** but cannot **open the website**, the most likely problem is at the **upper OSI layers**, especially **Layer 7 (Application)** and possibly **Layer 4 (Transport)**.

Possible OSI layers causing the problem

- **Application Layer (Layer 7) — Most likely**

Reason: Ping works, so basic IP connectivity to the server exists. But opening a website depends on web services such as **HTTP or HTTPS**. The web server may be down, misconfigured, or refusing requests.

Example: The server responds to ping, but its HTTP/HTTPS service is not working.

- **Transport Layer (Layer 4) — Also possible**

Reason: Websites normally use **TCP port 80 for HTTP** and **TCP port 443 for HTTPS**. These ports might be blocked by a firewall or the server may not be listening on them.

Example: Ping succeeds, but TCP port 443 is blocked.

- **Application Layer / DNS issue — if accessing by domain name**

Reason: If I can ping the IP address but type a website name such as `example.com` and it does not resolve, there may be a **DNS problem**. DNS is generally treated as an Application Layer protocol.

The likely problem is at the Application Layer (Layer 7) or Transport Layer (Layer 4).

Successful ping indicates that **Network Layer (Layer 3) connectivity is working**, while website access can still fail because of an **HTTP/HTTPS service problem, DNS issue, blocked TCP port 80/443, or web server configuration issue**.